Applied Study

Supplementary material for the CountBCF manuscript

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1 Applied Study

The simulation experiments establish the operating characteristics of CountBCF under controlled data-generating mechanisms. We now assess it in a setting where the outcome, exposure, and scientific benchmarks are all real. We analyse solid-cancer incidence in the Radiation Effects Research Foundation (RERF) Life Span Study (LSS) of survivors of the Hiroshima and Nagasaki atomic bombings. The LSS is the principal source of quantitative evidence on radiation-related cancer risk in humans, with detailed follow-up and well-characterised dosimetry supporting successive incidence analyses [Thompson et al., 1994, Preston et al., 2007, Grant et al., 2017].

The application has two purposes. First, it is a convergence check: on an outcome for which carefully constructed excess-relative-risk models are available, do the marginal effects recovered by CountBCF agree with the corresponding parametric analysis? Second, it is a substantive demonstration: can a nonparametric treatment-effect surface recover established variation in radiation risk without the analyst pre-specifying interactions in sex, age at exposure, or attained age? The answer to both questions is affirmative for the well-powered all-solid-cancer contrast. CountBCF also identifies one qualified departure from the standard model, a city difference in the low-dose contrast. That result is robust as an association but does not, on this evidence alone, identify a biological difference between cities.

This document is the supplementary version of the applied study and reports the complete set of tables, figures, sensitivity fits, and diagnostics. A condensed version appears in the main text. All analyses use the complete 1958–1998 person-year table; results are produced by the pipeline in `Applied_Study_v2/code/`, with tables written to `analysis/tables/` and figures to `analysis/plots/`.

1.1 Data validation against the published analysis

Before any modelling, the pipeline verifies that our handling of the public file reproduces the published totals. Table 1 compares our analysis frame with Table 4 of Preston et al. [2007]. Every case count and every head count matches exactly, and person-years agree to a maximum relative difference of 2.6×10^{-4}, which is attributable to rounding in the published table. Because the data-handling step is verified to the case, any disagreement in the results that follow is a modelling difference rather than a data-handling difference.

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Table 1: Validation of the analysis frame against Table 4 of Preston et al. [2007]. Case and person counts match exactly; the person-year differences reflect rounding in the published table.

Stratum	Cases	Published	Person-years	Published	People	Published
Not in city	3,994	3,994	680,744	680,744	25,247	25,247
$<$ 0.005 Gy	5,603	5,603	918,200	918,199	35,545	35,545
0.005–0.1	4,406	4,406	729,604	729,603	27,789	27,789
0.1–0.2	968	968	145,925	145,925	5,527	5,527
0.2–0.5	1,144	1,144	153,886	153,846	5,935	5,935
0.5–1	688	688	81,251	81,251	3,173	3,173
1–2	460	460	41,413	41,412	1,647	1,647
2–4	185	185	13,712	13,711	564	564
Male	7,969	7,969	1,040,281	1,040,278	42,902	42,902
Female	9,479	9,479	1,724,454	1,724,452	62,525	62,525
Hiroshima	12,931	12,931	1,967,602	1,967,601	73,430	73,430
Nagasaki	4,517	4,517	797,133	797,129	31,997	31,997
Total	17,448	17,448	2,764,735	2,764,730	105,427	105,427

The composition of the all-solid endpoint also required verification, because it determines which site-specific mechanisms can drive the aggregate result. The all-solid total decomposes cell by cell, with zero discrepancies, as the sum of oral cavity, digestive, respiratory, skin/bone/connective tissue, breast, female genital, male genital, urinary, renal pelvis and ureter, central nervous system, thyroid, other solid, and thymus. Two codebook features make this non-obvious: the urinary group (661 known-dose cases) omits renal pelvis and ureter (80 cases) even though the documentation places the latter under the urinary system, and thymus (12 cases) sits outside the respiratory group. Restoring both recovers 17,448 exactly. Group by group against Table 2 of Preston et al. [2007], oral cavity (277), digestive (10,052), respiratory (2,001), skin, bone and connective tissue (400), breast (1,082), female genital (1,457), male genital (420), central nervous system (281), thyroid (471), stomach (4,730), lung (1,759), colon (1,516) and liver (1,494) all match the published counts to the case. Within the skin, bone and connective tissue group, non-melanoma skin cancer contributes 330 cases and melanoma 17, again exactly reproducing the published figures. Non-melanoma skin cancer is therefore inside the all-solid aggregate, consistent with the published description of 17,448 first primary cancers including non-melanoma skin cancer. This matters because non-melanoma skin is one of the few sites whose ERR varies significantly with age at exposure in the published heterogeneity tests, making it a candidate driver of the aggregate's age-at-exposure shape. Section 1.9 tests that candidate directly.

1.2 Data, treatment design, and estimands

The source is the public 1958–1998 LSS solid-cancer incidence person-year table. Each row is a stratum defined by city, sex, age at exposure, attained age, and follow-up period; it records cancer cases and person-years at risk. The analytic endpoint is all first primary solid cancers. Its 17,448 cases among 105,427 cohort members reproduce the published totals in Preston et al. [2007], as documented in Section 1.1.

Unknown-dose records are excluded: 1,236 cells carrying 174,631 person-years and 1,197 cases. The primary analysis is restricted to in-city survivors and uses the standard DS02 weighted colon dose, gamma dose plus ten times neutron dose, expressed in Gy. We contrast a common reference group below 5 mGy against three exposure bands: low (5–$<$ 100 mGy), medium (100–$<$ 500 mGy),

and high (500–< 2000 mGy). Survivors at or above 2 Gy are excluded from the primary contrasts and restored in a sensitivity fit. Each band is fitted as a binary CountBCF analysis against the same reference group. This categorical construction makes the causal contrast explicit and does not impose a dose-response curve. One point of nomenclature deserves mention: the dose columns are weighted, adjusted, truncated DS02 organ doses in Gy despite the codebook labelling them mGy, whereas the dose-category cutpoints are in mGy.

Survivors not in either city are excluded from the primary contrast because their probability of receiving an in-city bomb dose is zero, so including them in the control arm of a treated-versus-control contrast would violate the overlap condition that CountBCF relies on. This is not an objection to their use in a regression that merely borrows their follow-up to pin down the baseline shape, which is the conventional LSS treatment. We therefore include them in a sensitivity analysis with a not-in-city indicator in the prognostic forest, the CountBCF analogue of the standard baseline adjustment, and compare the two fits in Section 1.9. The published analysis provides the benchmark for this check, reporting that dose parameters are virtually unchanged when the not-in-city group is excluded and that including it without baseline adjustment inflates the standardised ERR by about 8%.

The prognostic forest receives city, sex, age at exposure, attained age, and the estimated propensity score; the moderating forest receives the same substantive covariates. Calendar year, birth cohort, and time since exposure are deterministic functions of the two age variables, since the identity year = age − age at exposure + 1946 holds to the digit in this file, so they are not entered separately. City is placed in the moderator set deliberately. The canonical risk models put city in the baseline only and never allow it to modify the ERR, so a credible city effect on the conditional effect would be a modifier that the canonical model cannot express. Ground distance from the hypocentre is omitted because it is part of the dose-assignment mechanism rather than a pre-exposure confounder; a preliminary analysis confirmed that conditioning on it collapses overlap and reverses the sign of the estimate. The propensity-score adjustment follows the regularization-induced-confounding strategy of Hahn et al. [2020], Rosenbaum and Rubin [1983].

With person-years as an offset, $\exp(\mu_{f,i})$ is the reference-arm rate in cell i and $\tau_{f,i}$ is the log rate ratio. We report both an absolute and a relative conditional effect:

$$\mathrm{EAR}_i = \{\exp(\mu_{f,i} + \tau_{f,i}) - \exp(\mu_{f,i})\} \times 10^4, \qquad \mathrm{ERR}_i = \exp(\tau_{f,i}) - 1. \tag{1}$$

The first is an excess absolute rate in cases per 10^4 person-years; the second is the ERR used in LSS risk modelling. The standardised ERR is the ratio of person-year-weighted counterfactual rates, $(\sum_i w_i r_{1,i}) \,/\, (\sum_i w_i r_{0,i}) - 1$, rather than an average of cell-specific rate ratios. The distinction is quantitatively important, not pedantic: cells with a high ERR tend to be cells with a low baseline rate, so the average of ratios runs well above the ratio of averages. For high-dose all-solid cancer, averaging per-cell rate ratios would report a rate ratio of 1.84 against the standardised 1.49, creating a spurious disagreement with the parametric benchmark. Parametric g-computation returns the standardised quantity, so that is what we compare with the GLM rate ratio and with published case-weighted summaries, while the per-cell average is the conditional-effect average used in the subgroup separation tests [Hernán and Robins, 2006, Robins, 1986]. All summaries are computed on the posterior draws, so a subgroup contrast such as the difference between a conditional effect and the average effect carries a genuine credible interval rather than being a difference of two point estimates.

The causal interpretation is conditional on consistency, no interference, positivity, and conditional exchangeability given the available covariates. Dose resulted from location and shielding at the time of the detonations, not a behavioural treatment choice, but the study remains observational. In particular, the public table contains no individual smoking, dietary, or infection

information. The estimates should therefore be read as causal under these assumptions and as adjusted associations otherwise.

1.3 Descriptive structure of the analysis table

Table 2: Descriptive summary of the all-solid-cancer analysis table. The reference and three treatment bands are in-city groups. Rates are crude and reported before covariate adjustment. The final column is the marginal variance-to-mean ratio of the cell counts.

Dose band	Cells	Subjects	Person-years	Cases	Rate/10^4 PY	Mean dose (Gy)	Zero cells (%)	Var/mean
Not in city	1,285	25,247	680,744	3,994	58.7	0	44.1	8.10
Reference, 0–< 5 mGy	2,568	35,545	918,200	5,603	61.0	0.00096	47.2	5.37
Low, 5–< 100 mGy	5,973	27,789	729,604	4,406	60.4	0.0308	68.6	3.42
Medium, 100–< 500 mGy	7,598	11,462	299,812	2,112	70.4	0.2354	80.8	1.65
High, 500–< 2000 mGy	5,681	4,820	122,664	1,148	93.6	0.9216	83.9	1.34

Table 2 illustrates why adjustment is necessary. The low-dose crude rate of 60.4 per 10^4 person-years is slightly below the reference rate of 61.0, although the adjusted dose response rises across the medium and high bands. This is confounding rather than biology. Age at exposure, attained age, and sex are almost perfectly balanced across bands, with standardised differences between 0.002 and 0.047, whereas city is not: Hiroshima accounts for 58% of the reference group and 78% of the low-dose group, giving standardised differences between 0.19 and 0.56. City is therefore the one covariate on which the propensity model has real work to do, and it is why the adjusted analyses move the low-band estimate.

The cells are sparse but not, on inspection, genuinely overdispersed. Roughly 74% of the analysis cells are zero and the marginal variance-to-mean ratio runs from 1.3 to 5.4. Once a flexible baseline is conditioned on, consisting of sex-specific splines in log attained age and in age at exposure plus a city term, the extra-Poisson variation essentially disappears: the fitted negative-binomial size is $\theta \approx 100$, implying a variance-to-mean ratio of 1.006 at the mean fitted rate; the negative-binomial AIC is no better than the Poisson AIC; and the likelihood-ratio test for the negative binomial over the Poisson gives $p = 0.5$. The quasi-Poisson scale, at a Pearson χ^2/df of 1.69, is inflated mostly by the sparse-cell behaviour of the Pearson statistic, since the deviance per degree of freedom is 0.59. The apparent overdispersion in this table is thus baseline heterogeneity rather than extra-Poisson noise, and a model flexible enough in the baseline absorbs it. We nevertheless run CountBCF with the negative-binomial likelihood because it nests the Poisson and lets the data decide; its posterior for the dispersion, reported in Section 1.5, reaches the same conclusion from the same fit that produces the causal estimates. The many zero cells are sampling zeros in thin person-year strata rather than evidence of a not-at-risk subpopulation.

The crude data already display the canonical gradient, which is the pattern against which any model output must be read. Comparing the high band to the reference within age-at-exposure strata, the crude rate ratio runs 3.41, 1.90, 1.51, 1.35, and 1.41 across age-at-exposure groups 0–9, 10–19, 20–29, 30–39, and 40 or more for women, and 1.71, 1.72, 1.41, 1.12, and 1.20 for men.

1.4 Parametric anchors

Three parametric anchors are fitted on the identical in-city analytic set: a refit of the canonical excess-relative-risk and excess-absolute-risk models, a fully categorical dose response, and adjusted generalised linear models for the marginal contrasts.

1.4.1 Refitting the canonical risk model

The standard model is refitted by maximum Poisson likelihood on the same in-city analysis set, using a sex-specific spline baseline in log attained age and in age at exposure plus a city term, with

$$\mathrm{ERR}(d, e, a, s) = \beta_s \, d \, \exp\{\alpha_e (e - 30) + \alpha_a \log(a/70)\}\,, \tag{2}$$

and the additive analogue for the EAR, where d is weighted colon dose, e is age at exposure, a is attained age, and s is sex. Because this refit is the comparator against which the CountBCF surface is measured in Section 1.7, it matters that it reproduces the published fit. Table 3 shows that every one of the ten published parameters is recovered inside its published 90% confidence interval. The parametric yardstick is therefore the established risk model itself, rather than a simplified regression surrogate.

Table 3: Refit of the canonical ERR and EAR risk models on the in-city analytic set, compared with Table 10 of Preston et al. [2007]. Intervals are 90% confidence intervals. Every refitted parameter falls inside the published interval.

Parameter	Refit (90% CI)	Published (90% CI)
ERR per Gy, male, attained age 70, exposed at 30	0.336 (0.245, 0.426)	0.35 (0.28, 0.43)
ERR per Gy, female, attained age 70, exposed at 30	0.554 (0.438, 0.670)	0.58 (0.43, 0.69)
ERR female-to-male ratio	1.65	1.6 (1.31, 2.09)
ERR change per decade of age at exposure	−15.2% (−25.6%, −3.1%)	−17% (−25%, −7%)
ERR attained-age power	−1.87 (−2.45, −1.28)	−1.65 (−2.1, −1.2)
EAR per 10^4 PY-Gy, male	38.9 (25.8, 52.1)	43 (33, 55)
EAR per 10^4 PY-Gy, female	57.8 (47.6, 68.1)	60 (51, 69)
EAR female-to-male ratio	1.49	1.4 (1.10, 1.79)
EAR change per decade of age at exposure	−23.5%	−24% (−32%, −16%)
EAR attained-age power	+2.25 (1.73, 2.77)	2.38 (1.9, 2.8)

Fitting one parameter per dose category on top of the same baseline gives 18 nonparametric dose points. A weighted linear fit through the origin gives an ERR of 0.61 per Gy with a standard error of 0.04 in the absence of effect modification. This is larger than the 0.47 standardised to attained age 70 and exposure at age 30, as expected, since the unstandardised average is taken over a cohort whose younger-exposed members carry higher relative risk. The category estimates rise steadily and give no visible sign of a threshold (Figure 1).

1.4.2 Generalised linear model marginal effects

Table 4 reports g-computation marginal effects from adjusted Poisson and negative-binomial models. The Poisson and negative-binomial fits are nearly indistinguishable for all-solid cancer, consistent with the dispersion diagnostics of Section 1.3. Adding dose by modifier interactions improves the all-solid negative-binomial fit by $\Delta\mathrm{AIC} = 54$ (26,931 against 26,985), so heterogeneity is detectable parametrically once the analyst knows to look for it.

Two features of the parametric family are worth carrying forward. The first is the improvement just noted, which establishes that the heterogeneity CountBCF reports in Section 1.6 is present in the data and not a property of the forest. The second is that the unadjusted models are unusable: a minimal negative-binomial model for high-dose stomach returns a rate ratio of 63 with a 95% interval of (0.08, 521), and a minimal negative-binomial model for low-dose all-solid cancer flips to a rate ratio of 0.90 against the Poisson's 0.99. The parametric answer therefore depends entirely

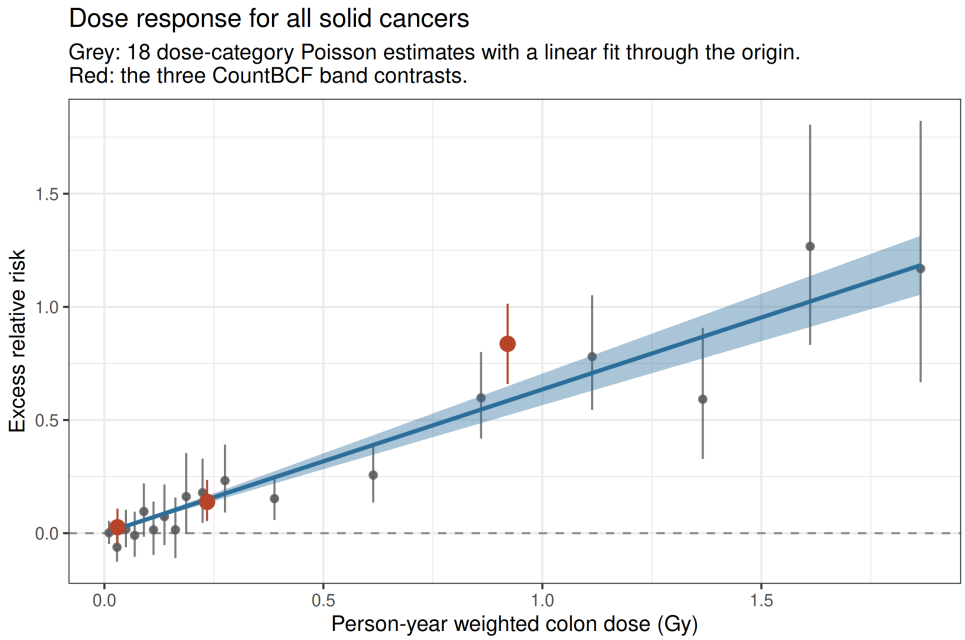


Figure 1: Categorical-dose response from the adjusted parametric model. The estimates increase with dose and do not suggest a threshold over the range considered.

on which likelihood and which covariate set the analyst happens to choose, which is precisely the fragility that motivates the present method.

Table 4: Marginal effects from adjusted parametric models by g-computation. Excess rates are cases per 10^4 person-years. Intervals are 95% confidence intervals.

Endpoint	Contrast	Model	Excess rate (95% CI)	Rate ratio (95% CI)
All solid	High	Poisson (adjusted)	32.6 (26.8, 38.3)	1.532 (1.435, 1.632)
	High	NB (adjusted)	32.7 (27.1, 38.5)	1.535 (1.440, 1.635)
	Medium	NB (adjusted)	7.35 (3.96, 10.70)	1.120 (1.064, 1.178)
	Low	NB (adjusted)	−0.11 (−2.60, 2.40)	0.998 (0.958, 1.040)
Stomach	High	NB (adjusted)	5.30 (2.57, 8.14)	1.319 (1.152, 1.495)
	Medium	NB (adjusted)	1.03 (−0.67, 2.80)	1.063 (0.961, 1.173)
	Low	NB (adjusted)	0.30 (−1.05, 1.62)	1.019 (0.940, 1.100)

1.5 Average effects and agreement with parametric risk models

Each contrast is fitted with two chains, 2,000 burn-in and 1,000 retained draws per chain. The Gelman–Rubin statistic is 1.00 for every contrast, with a maximum of 1.02, and effective sample sizes run from 276 to 1,617 for the exposed-arm effect and from 708 to 1,561 for the ERR. The negative-binomial dispersion posterior is $\theta \approx 157$–190 for all-solid cancer (95% credible interval 56 to 585) and $\theta \approx 21$–30 for stomach (95% credible interval 8.5 to 77). The all-solid cell counts are thus effectively Poisson once the prognostic forest has done its work, while stomach retains mild genuine overdispersion. This is a sharper answer to the overdispersion question than the generalised-linear-model diagnostics provide, and it is obtained from the same fit that produces the

causal estimates.

CountBCF and the adjusted negative-binomial generalised linear model agree in every marginal contrast (Table 5 and Figure 2). For high-dose all-solid cancer, CountBCF estimates an excess of 30.2 cases per 10^4 person-years and a rate ratio of 1.488, compared with 32.7 and 1.535 from the parametric model. The medium-dose contrast is positive under both methods, whereas the low-dose band is uninformative as an overall contrast. CountBCF lies slightly nearer the null throughout, consistent with regularisation of the treatment-effect surface. The exposed-arm version of the estimand is within 0.5% of the population-average version throughout, at 30.37 against 30.22 per 10^4 person-years for high-dose all-solid cancer, and it is the exposed-arm version that Sections 1.7 and 1.14 use, since a per-Gy risk and an attributable fraction are properties of the exposed.

Table 5: Marginal effects from CountBCF and the adjusted negative-binomial GLM. Excess rates are cases per 10^4 person-years and ERR equals the rate ratio minus one. Intervals are 95% credible intervals for CountBCF and 95% confidence intervals for the GLM.

Endpoint	Contrast	Method	Excess rate (95% interval)	ERR (95% interval)	Rate ratio (95% interval)
All solid	High	CountBCF	30.2 (23.9, 35.9)	0.488 (0.381, 0.584)	1.488 (1.381, 1.584)
	High	NB GLM	32.7 (27.1, 38.5)	0.535 (0.440, 0.635)	1.535 (1.440, 1.635)
	Medium	CountBCF	6.95 (3.22, 10.67)	0.112 (0.051, 0.173)	1.112 (1.051, 1.173)
	Medium	NB GLM	7.35 (3.96, 10.70)	0.120 (0.064, 0.178)	1.120 (1.064, 1.178)
	Low	CountBCF	−0.32 (−2.90, 2.26)	−0.005 (−0.046, 0.037)	0.995 (0.954, 1.037)
	Low	NB GLM	−0.11 (−2.60, 2.40)	−0.002 (−0.042, 0.040)	0.998 (0.958, 1.040)
Stomach	High	CountBCF	4.28 (1.64, 7.11)	0.252 (0.095, 0.426)	1.252 (1.095, 1.426)
	Medium	CountBCF	0.22 (−1.58, 2.09)	0.014 (−0.087, 0.125)	1.014 (0.913, 1.125)
	Low	CountBCF	−0.59 (−2.03, 0.87)	−0.033 (−0.111, 0.051)	0.967 (0.889, 1.051)

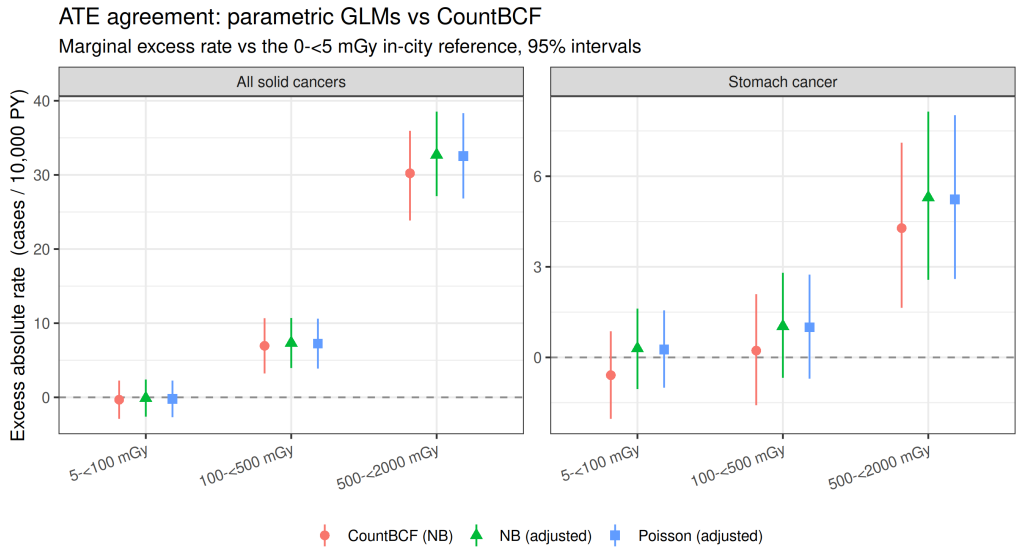


Figure 2: Convergence of marginal effects from CountBCF and the adjusted parametric model. Each estimate is a person-year-standardised excess rate with a 95% interval.

The high-dose band has a mean dose contrast of 0.9207 Gy relative to the reference group. Thus, the standardised high-dose ERR of 0.488 (0.381, 0.584) in Table 5, divided by 0.9207, gives 0.529 per Gy (95% credible interval 0.419, 0.630), reported as 0.53 (0.42, 0.63) per Gy. This is a descriptive rescaling of the categorical high-versus-reference contrast, not a separately estimated

continuous-dose slope. On this common per-Gy scale it is close to the published standardised ERR of 0.47 (0.40, 0.54) from Preston et al. [2007]. The medium band gives 0.39 (0.15, 0.65) per Gy on the same rescaling, and for stomach the high band gives 0.26 (0.10, 0.44) per Gy against the published 0.34. The dose response implied by the three band contrasts is consistent with linearity over 0 to 2 Gy: the standardised ERRs of −0.019, 0.092, and 0.487 sit at mean dose contrasts of 0.030, 0.234, and 0.921 Gy, implying per-Gy slopes of roughly −0.63, 0.39, and 0.53, with the low band uninformative because its dose contrast is only 30 mGy (Figure 1). A model given no linearity constraint thus recovers the principal average pattern in the canonical analysis.

Table 6 gives the corresponding E-value sensitivity analysis [VanderWeele and Ding, 2017]. For the high-dose all-solid-cancer contrast, an unmeasured confounder would need associations of at least 2.34 with both dose-band membership and cancer incidence, on the rate-ratio scale, to explain the point estimate away; associations of at least 2.12 with both would be needed to move the interval limit nearest the null to the null. These values are not a proof that unmeasured confounding is absent, but they indicate that a confounder would need to be materially stronger than one weakly related to location-assigned dose. Given that the arms are balanced on every measured demographic except city, and that exposure was determined by where a person happened to be standing on the morning of the bombing, confounders of that strength are hard to name. The medium-dose all-solid result is less robust (E-values 1.41 and 1.23), and the high-dose stomach result is intermediate (1.81 and 1.42). For the low-dose contrasts, the interval E-value is one because the corresponding interval already contains the null. The E-values therefore reinforce the evidence for the high-dose all-solid association while leaving low-dose and site-specific inferences appropriately more tentative.

Table 6: E-value sensitivity analysis for CountBCF. The rate ratios are exposed-arm, person-year-standardised estimates (ATT), which differ slightly from the population-average rate ratios in Table 5. The point and interval E-values give the minimum strength of association, on the rate-ratio scale, that an unmeasured confounder would need with both treatment and outcome to explain the point estimate or move the interval nearest the null to the null.

Endpoint	Contrast	Exposed-arm RR (95% CrI)	E-value, point	E-value, interval
All solid	Low	0.981 (0.940, 1.023)	1.16	1.00
	Medium	1.092 (1.035, 1.152)	1.41	1.23
	High	1.487 (1.386, 1.580)	2.34	2.12
Stomach	Low	0.956 (0.877, 1.040)	1.27	1.00
	Medium	1.002 (0.900, 1.112)	1.04	1.00
	High	1.252 (1.095, 1.426)	1.81	1.42

1.6 Heterogeneity of the radiation-effect surface

The main advantage of CountBCF in this application is not a different average effect, but its ability to estimate effect modification without pre-specifying the interaction functions. Throughout this section we call a subgroup difference credible only when the posterior interval for its person-year-weighted mean ERR minus the cohort-wide average ERR (the standardised ATE) excludes zero. The EAR is reported alongside it to describe the absolute burden, but it does not determine the separation indicator. This guards against treating a difference in posterior means as evidence of genuine heterogeneity.

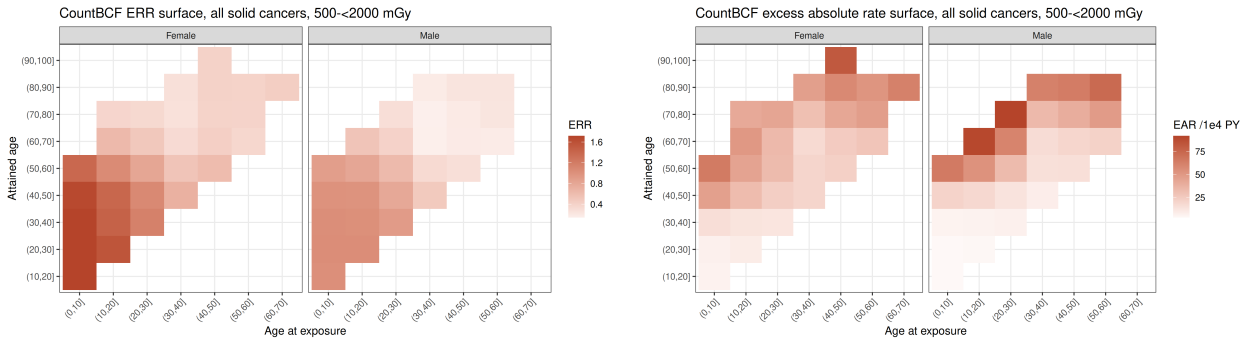
1.6.1 How much heterogeneity is present at all

Before examining named subgroups it is useful to ask how much variation the posterior surface contains. Table 7 reports the posterior person-year-weighted standard deviation of the per-cell ERR across exposed cells, together with the share of exposed cells whose effect is positive and the share whose 95% interval excludes zero.

Table 7: Cell-level heterogeneity of the CountBCF effect surface. The first column is the posterior person-year-weighted standard deviation of the per-cell ERR across exposed cells.

Endpoint	Contrast	SD of per-cell ERR (95% CrI)	Cells with ERR > 0	Cells with CrI excluding 0
All solid	High	0.584 (0.416, 0.773)	98.6%	63.0%
	Medium	0.253 (0.198, 0.324)	82.8%	2.0%
	Low	0.233 (0.183, 0.299)	58.1%	2.4%
Stomach	High	0.004 (0.000, 0.030)	100%	100%
	Medium	0.146 (0.076, 0.240)	73.5%	0%
	Low	0.186 (0.135, 0.257)	54.3%	0%

For all-solid cancer at high dose the heterogeneity is large and credible: the per-cell ERR has a posterior median of 0.60 with a tenth-to-ninetieth percentile range of 0.16 to 1.57, and nearly two thirds of individual cells have an interval excluding zero. At medium and low dose the surface is much flatter and almost no individual cell separates, which is a statement about power rather than about the absence of modification. For stomach at high dose the moderating forest returns an essentially constant effect, an ERR of 0.252 in every subgroup with a posterior standard deviation of 0.004; that case is discussed in Section 1.6.4. Figures 3a and 3b display the fitted relative and absolute surfaces over the covariate space.



(a) Relative effect (ERR). (b) Absolute effect (EAR).

Figure 3: Estimated CountBCF effect surfaces for high-dose all-solid cancer over the observed covariate space. The relative and absolute surfaces move in opposite directions with attained age.

1.6.2 All solid cancers at 500–< 2000 mGy

Table 8 gives the complete set of high-dose subgroup effects with their separation tests. Five conclusions follow.

First, age at exposure is the dominant modifier of the relative effect. The ERR declines more than fourfold, from 1.45 (0.97, 1.98) among those exposed at ages 0–9 to 0.32 (0.16, 0.48) among

those exposed at age 40 or later, and both tails separate credibly from the average. This is the textbook LSS result, recovered with no interaction term specified anywhere.

Second, attained age modifies the relative and absolute effects in opposite directions. The ERR falls from 1.41 (0.97, 1.89) below age 40 to 0.32 (0.13, 0.54) at age 80 or older, while the EAR rises from 8.7 to 57.3 excess cases per 10^4 person-years, and every attained-age band separates. The two are projections of a single surface, since the EAR is approximately the baseline rate multiplied by the ERR and the baseline rate climbs faster with attained age than the relative effect fades.

Third, time since exposure moves the absolute effect but not the relative one. The ERR is statistically flat across 20 to more than 40 years of follow-up, with no subgroup separating, while the EAR nearly quadruples from 15.2 to 57.3 per 10^4 person-years. The relative excess established early does not wash out; the absolute burden grows because the baseline does.

Fourth, sex does not separate at high dose, although the direction is now the canonical one, with a female ERR of 0.94 against a male 0.74, a ratio of 1.27 in the same direction as the published 1.6. This reverses a finding from a preliminary analysis on a 30% subsample, which had the male ERR marginally above the female one. Section 1.8 shows that the sex difference is also sensitive to treatment-effect regularisation, and Section 1.9 shows that it is carried by breast and thyroid, so we do not treat it as a primary heterogeneity finding.

Fifth, the two cities are indistinguishable at high dose, which is what one would want if the DS02 weighted organ doses have reconciled the Hiroshima neutron component correctly.

Table 8: High-dose all-solid-cancer conditional effects (500–< 2000 mGy against the 0–< 5 mGy reference). A subgroup is marked as separated when the 95% posterior interval for its person-year-weighted mean ERR minus the cohort-wide average ERR (the standardised ATE) excludes zero. The Cases column counts cases in the subgroup across both dose arms, and the EAR column is descriptive; neither is used for this classification.

Dimension	Subgroup	Cases	ERR (95% CrI)	CATE − ATE (95% CrI)	Separates?
Sex	Male	3,089	0.74 (0.49, 1.05)	−0.12 (−0.31, 0.07)	No
	Female	3,662	0.94 (0.73, 1.18)	+0.07 (−0.04, 0.19)	No
Age at exposure	0–9	532	1.45 (0.97, 1.98)	+0.58 (0.24, 0.96)	Yes
	10–19	1,303	1.08 (0.81, 1.38)	+0.21 (0.04, 0.42)	Yes
	20–29	1,159	0.72 (0.49, 0.98)	−0.15 (−0.38, 0.09)	No
	30–39	1,499	0.37 (0.17, 0.58)	−0.50 (−0.74, −0.27)	Yes
	40+	2,258	0.32 (0.16, 0.48)	−0.55 (−0.79, −0.32)	Yes
Attained age	< 40	167	1.41 (0.97, 1.89)	+0.54 (0.27, 0.87)	Yes
	40–59	1,677	1.00 (0.77, 1.25)	+0.13 (0.02, 0.24)	Yes
	60–79	3,818	0.36 (0.24, 0.49)	−0.51 (−0.71, −0.31)	Yes
	80+	1,089	0.32 (0.13, 0.54)	−0.55 (−0.82, −0.27)	Yes
City	Hiroshima	4,369	0.77 (0.57, 0.97)	−0.10 (−0.27, 0.04)	No
	Nagasaki	2,382	1.02 (0.69, 1.39)	+0.15 (−0.06, 0.39)	No
Time since exposure	< 20 y	939	0.93 (0.67, 1.18)	+0.06 (−0.04, 0.17)	No
	20–29 y	1,430	0.90 (0.68, 1.13)	+0.03 (−0.02, 0.09)	No
	30–39 y	1,670	0.88 (0.69, 1.08)	+0.01 (−0.04, 0.06)	No
	40+ y	2,712	0.75 (0.57, 0.94)	−0.12 (−0.27, 0.03)	No

Excess absolute rates per 10^4 person-years for the same subgroups, in the order listed, are 28.9 and 31.0 by sex; 24.6, 34.3, 31.8, 24.4, and 35.5 by age at exposure; 8.7, 34.3, 37.6, and 57.3 by attained age; 33.0 and 26.2 by city; and 15.2, 20.8, 32.1, and 57.3 by time since exposure.

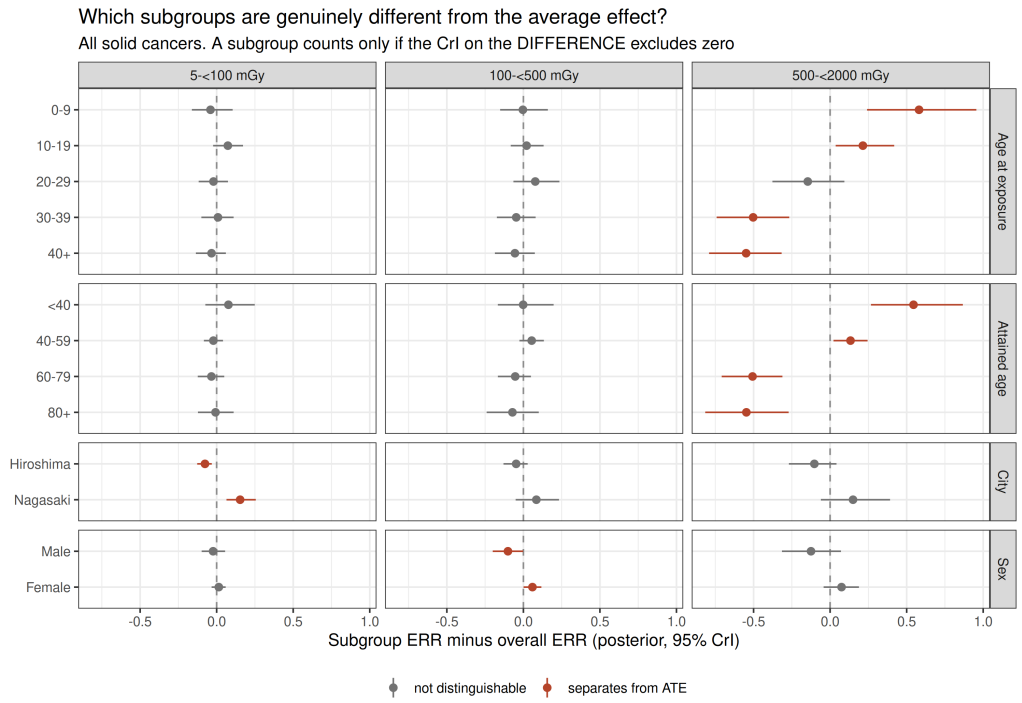


Figure 4: Subgroup separation for the high-dose all-solid contrast. Each point is the posterior mean of the subgroup conditional effect minus the population average effect (ATE), with its 95% credible interval. Intervals excluding zero identify credible effect modification.

These patterns match the central qualitative findings of LSS incidence modelling: relative risk is greater following younger-age exposure, while the absolute rate burden grows at older attained ages [Preston et al., 2007, Grant et al., 2017].

1.6.3 All solid cancers at 100–< 500 mGy

At medium dose the sex contrast is the one dimension that separates. The male ERR is 0.061 (−0.066, 0.214), with a difference from the average of −0.101 (−0.200, −0.002), while the female ERR is 0.222 (0.112, 0.348), with a difference of +0.059 (0.001, 0.118). No age-at-exposure or attained-age subgroup separates at this dose. The sex effect on the relative risk is thus credibly present at 100–< 500 mGy and credibly not resolvable at 500–< 2000 mGy, where the age-at-exposure structure dominates the moderating forest's budget. We note in Section 1.15 that this medium-dose sex effect has not been put through the regularisation sweep and inherits the doubt attached to its high-dose counterpart until it has been.

1.6.4 Stomach cancer as a counterpoint

Stomach cancer provides a useful counterpoint. Its high-dose average effect is positive, but its estimated ERR surface is essentially constant across the examined covariates, with a posterior standard deviation of 0.004. This is not a failure of the estimator; it is the differential-shrinkage prior doing what it is designed to do. With 269 exposed stomach cases the data cannot support effect modification, and the published tests agree, giving $P = 0.4$ for the stomach age-at-exposure trend and $P = 0.05$ for the attained-age effect. Where the data do not support a complex surface,

the fit shrinks toward a homogeneous effect rather than creating spurious subgroup differences. Whether that shrinkage is too aggressive is exactly what Section 1.8 tests for the all-solid endpoint, and Section 1.15 records that the sweep was not repeated for stomach.

1.7 Comparison with the canonical LSS surface

The average-effect comparison is only a partial validation. We therefore evaluate the CountBCF surface against the refitted risk model of Section 1.4.1 at the same person-year cells. Two comparisons are possible. The standardised comparison weights each surface by the same CountBCF-estimated control-arm rate, which makes it the analogue of a rate ratio and the fair test. The per-cell comparison is the plain person-year-weighted mean of the cell ERRs; it is systematically larger for both surfaces and is what a naive comparison would produce. We report the standardised comparison. Unlike the separation tests of Section 1.6, which standardise the conditional effect over the whole cohort, this comparison is evaluated only at the exposed person-year cells, because the parametric ERR is proportional to dose and collapses to zero at the reference cells whereas the CountBCF contrast does not; the two surfaces can therefore only be compared where the exposure actually occurred. A given subgroup's CountBCF ERR here can accordingly differ by a few percent from its cohort-standardised value in Table 8.

1.7.1 All solid cancers

On the standardised scale the two surfaces agree in all but two of 28 high-dose subgroups and all but three of 28 medium-dose subgroups (Table 9 and Figure 5). Their person-year-weighted mean EAR is identical to four significant figures at high dose, 30.37 per 10^4 person-years for each model, with a rank correlation of 0.57 across the 5,681 exposed cells. The agreement is notable because CountBCF was supplied neither a linear dose term nor any interaction involving age, sex, or city. A nonparametric causal forest, given no functional form and no interaction terms, reproduces the canonical parametric risk surface for all solid cancers over almost the whole covariate space.

Table 9: Standardised comparison of the CountBCF surface with the refitted canonical risk model, evaluated at the same person-year cells. Differences whose 95% credible interval excludes zero are the departures discussed in the text.

Contrast	Subgroup	Refitted model ERR	CountBCF ERR (95% CrI)	Difference (95% CrI)
High	Overall	0.553	0.487 (0.386, 0.580)	−0.066 (−0.168, 0.026)
	Attained age < 40	2.759	1.330 (0.931, 1.758)	−1.429 (−1.863, −0.965)
	Time since exposure < 20 y	0.659	0.471 (0.301, 0.640)	−0.188 (−0.356, −0.015)
	Age at exposure 0–9	1.522	1.184 (0.794, 1.607)	−0.338 (−0.735, 0.090)
	Female	0.707	0.603 (0.465, 0.741)	−0.104 (−0.243, 0.035)
	Male	0.396	0.369 (0.240, 0.500)	−0.027 (−0.157, 0.104)
Medium	Overall	0.138	0.092 (0.035, 0.152)	−0.046 (−0.103, 0.013)
	Age at exposure 0–9	0.368	0.114 (−0.053, 0.294)	−0.253 (−0.421, −0.073)
	Attained age < 40	0.736	0.142 (−0.072, 0.389)	−0.593 (−0.813, −0.340)

The disagreements are localized rather than scattered. They occur at young attained age and short time since exposure, where the parametric attained-age power law extrapolates most strongly. At attained age below 40 the refitted model predicts a high-dose ERR of 2.76 at 0.92 Gy, while CountBCF estimates 1.33 (0.93, 1.76) and excludes the parametric value. The same pattern appears one dose band down, at 0.74 against 0.14, and for stomach, at 2.89 against 0.25. A power law in attained age is forced to be monotone and unbounded as attained age tends to zero; the data in

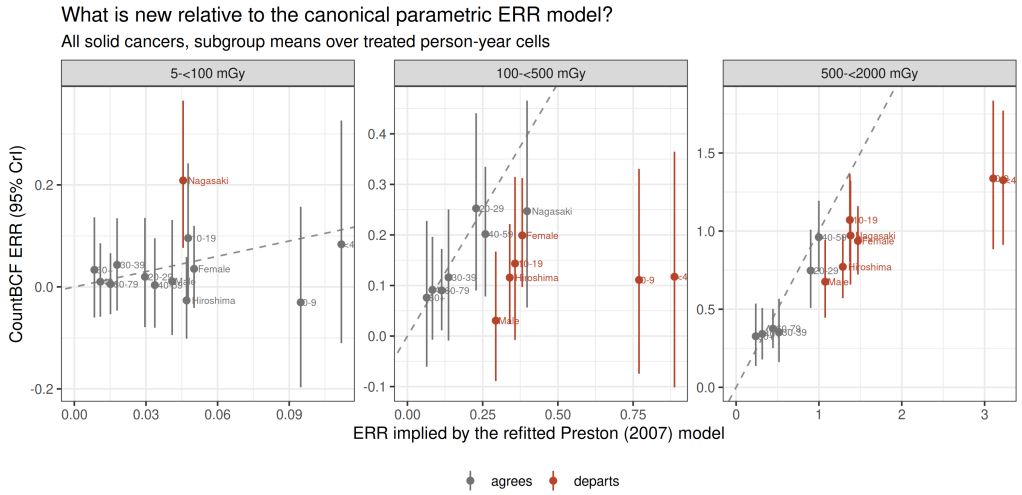


Figure 5: CountBCF conditional ERRs compared with the refitted canonical risk model. The models agree across most of the observed covariate space; the largest differences are confined to young attained age and short latency.

that corner are sparse, with only 51 exposed cases below attained age 40 in the high band, and CountBCF's answer is a flattening rather than a divergence. The published analysis itself flags this region as unsettled, noting that a more complex pattern provides a somewhat better fit for all solid cancers and that future analyses should give more attention to alternative descriptions of age-at-exposure effects on both the ERR and the EAR [Preston et al., 2007]. The competing explanation has to be stated plainly: because the moderating-forest prior shrinks toward a constant, a flattening in a thin corner is also what excessive shrinkage would produce. Section 1.8 is the decisive test of that alternative.

1.7.2 Stomach cancer

For stomach at high dose, 16 of 23 subgroups depart on the standardised scale, because the parametric model carries effect modification, with an ERR proportional to attained age to the power -1.5, a decline of 13% per decade of age at exposure, and a female-to-male ratio of 2.3, while CountBCF returns a flat ERR of 0.252. The two are not equally supported by the data. The published significance tests for stomach give $P = 0.4$ for the age-at-exposure trend and $P = 0.05$ for attained age, so the modification being fitted is at or beyond the edge of detectability and its estimates are borrowed from the functional form rather than read off the stomach data. CountBCF, which must find modification in the stomach data alone, finds none. The right reading is not that either analysis is wrong about stomach, but that the apparent stomach effect modification in the canonical analysis is a property of the model class rather than of the stomach data. What the discrepancy identifies is the assumption doing the work: CountBCF requires the site-specific data to support a modifier, whereas the parametric estimate extrapolates a specified functional form.

1.8 Robustness to regularisation of the moderating forest

Because CountBCF deliberately tightens the prior on the treatment-effect surface in order to suppress spurious heterogeneity, any featured subgroup effect must be shown to survive both loosening and tightening that prior. Table 10 reports twelve refits of the featured contrast, all solid can-

cers at 500–< 2000 mGy, varying the number of moderating trees, the tree-depth penalty on the moderating forest, and the overall leaf concentration. Figure 6 displays the same sweep.

Table 10: Sensitivity of the featured contrast (all solid cancers, 500–< 2000 mGy) to regularisation of the moderating forest. Entries in the last two columns are posterior differences with 95% credible intervals.

Specification	Standardised ERR (95% CrI)	ERR, exposed < 20 minus 40+	ERR, female minus male
$m_\tau = 10$ (tightest)	0.434 (0.345, 0.521)	0.475 (0.333, 0.868)	0.098 (−0.102, 0.284)
$m_\tau = 25$	0.486 (0.389, 0.582)	0.772 (0.450, 1.110)	0.239 (−0.011, 0.502)
$m_\tau = 50$ (default)	0.495 (0.401, 0.594)	0.795 (0.447, 1.139)	0.311 (0.004, 0.607)
$m_\tau = 100$	0.508 (0.416, 0.608)	0.900 (0.560, 1.289)	0.293 (−0.021, 0.618)
$m_\tau = 200$ (loosest)	0.505 (0.405, 0.605)	0.982 (0.621, 1.362)	0.399 (0.053, 0.757)
Depth prior (0.05, 4), tight	0.494 (0.400, 0.599)	0.834 (0.487, 1.219)	0.199 (−0.065, 0.455)
Depth prior (0.25, 3), default	0.495 (0.401, 0.594)	0.795 (0.447, 1.139)	0.311 (0.004, 0.607)
Depth prior (0.5, 2), loose	0.496 (0.398, 0.598)	0.812 (0.480, 1.159)	0.260 (−0.039, 0.555)
Depth prior (0.95, 2), prognostic-like	0.489 (0.397, 0.585)	0.872 (0.488, 1.263)	0.299 (−0.022, 0.583)
$a_0 \times 0.5$	0.465 (0.359, 0.576)	0.475 (0.215, 0.746)	−0.034 (−0.072, 0.027)
$a_0 \times 1$ (default)	0.495 (0.401, 0.594)	0.795 (0.447, 1.139)	0.311 (0.004, 0.607)
$a_0 \times 2$	0.463 (0.367, 0.558)	1.301 (0.673, 2.096)	0.315 (−0.251, 0.824)

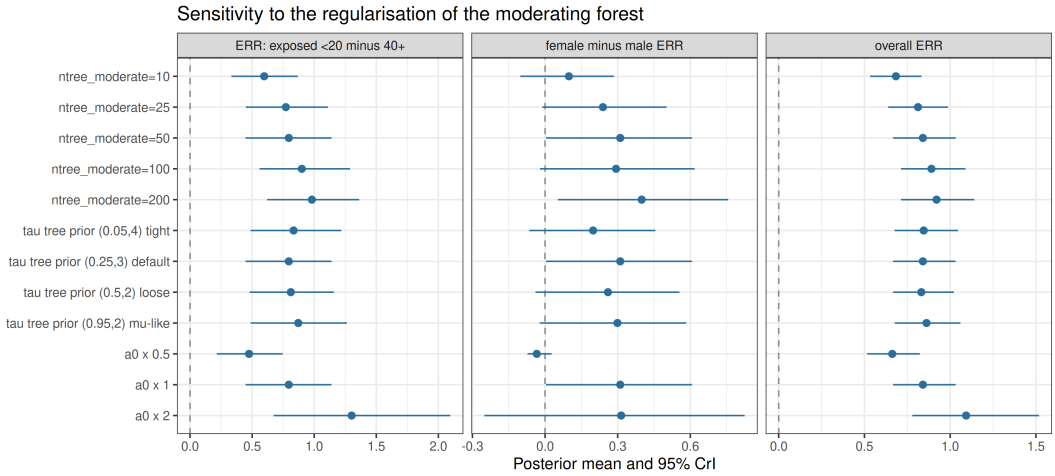


Figure 6: Sensitivity of the featured contrast to the moderating-forest prior. The overall-effect panel plots the person-year-weighted mean of the per-cell ERR, which differs from the standardised ERR tabulated in Table 10 for the reason given in Section 1.2.

Three conclusions follow. First, the average effect is essentially prior-free: the standardised ERR sits between 0.434 and 0.508 across a twentyfold change in the prior variance of the moderating surface and across every depth prior tried, so the headline number is not an artefact of the shrinkage. Second, the age-at-exposure gradient survives everything. The ERR difference between those exposed before age 20 and after age 40 has a credible interval excluding zero in all twelve specifications, including the tightest prior the sampler will accept, which gives 0.475 (0.333, 0.868), and the halved leaf concentration, which gives 0.475 (0.215, 0.746). Its magnitude scales with prior looseness, from 0.48 at the tightest to 1.30 at the loosest, so the size of the gradient is prior-dependent while its existence is not. Third, the sex difference does not survive. It is 0.311 (0.004, 0.607) at the defaults, but the interval covers zero at three of the five moderating-tree settings, at three of the four depth priors, and, most tellingly, it moves to −0.034 (−0.072, 0.027) when

the leaf concentration is halved. The sex effect is present only where the prior is loose enough to permit it and should not be featured. Finally, the Hiroshima-minus-Nagasaki ERR difference never separates from zero in any of the twelve specifications at high dose, confirming the null reported in Section 1.6.2.

1.9 Robustness to the data construction

Table 11 reports fourteen refits of the featured contrast, each changing one element of the data construction, with Figure 7 showing the same fits.

Table 11: Sensitivity of the featured contrast to the data construction. Each row changes one element relative to the baseline. The last column is the posterior ERR difference between those exposed before age 20 and after age 40, with its 95% credible interval.

Perturbation	Cells	Cases	Standardised ERR (95% CrI)	EAR/10^4 PY	ERR, $<$ 20 minus 40+
Baseline (all solid, colon bands, NB)	8,249	6,751	0.495 (0.401, 0.594)	30.7	0.795 (0.447, 1.139)
Poisson likelihood	8,249	6,751	0.498 (0.402, 0.598)	30.8	0.827 (0.484, 1.205)
Not in city added to control arm	9,534	10,745	0.501 (0.408, 0.603)	30.9	0.887 (0.555, 1.256)
Reference widened to 0–$<$ 100 mGy	14,222	11,157	0.503 (0.406, 0.606)	31.0	0.806 (0.520, 1.144)
High band extended to include $\geq$ 2 Gy	10,714	6,936	0.588 (0.494, 0.691)	35.9	1.110 (0.732, 1.520)
Excluding breast and thyroid	8,249	6,137	0.405 (0.310, 0.510)	23.1	0.350 (0.069, 0.636)
Excluding non-melanoma skin	8,249	6,605	0.479 (0.384, 0.576)	29.0	0.783 (0.440, 1.176)
Excluding stomach	8,249	4,975	0.575 (0.458, 0.697)	25.9	0.700 (0.332, 1.073)
Non-sex-specific solid only	8,249	5,502	0.525 (0.416, 0.642)	26.5	0.629 (0.285, 0.996)
Stomach, colon-dose bands	8,249	1,776	0.253 (0.078, 0.429)	4.41	−0.0001 (−0.007, 0.007)
Stomach, stomach-dose bands	8,075	1,774	0.256 (0.092, 0.443)	4.45	−0.0003 (−0.009, 0.006)
Lung, lung-dose bands	7,636	683	0.666 (0.356, 1.001)	4.21	−0.110 (−0.564, 0.265)
Colon, colon-dose bands	8,249	584	0.297 (0.019, 0.586)	1.64	0.034 (−0.233, 0.342)
Liver, liver-dose bands	7,917	592	0.291 (0.024, 0.609)	1.66	0.047 (−0.242, 0.376)

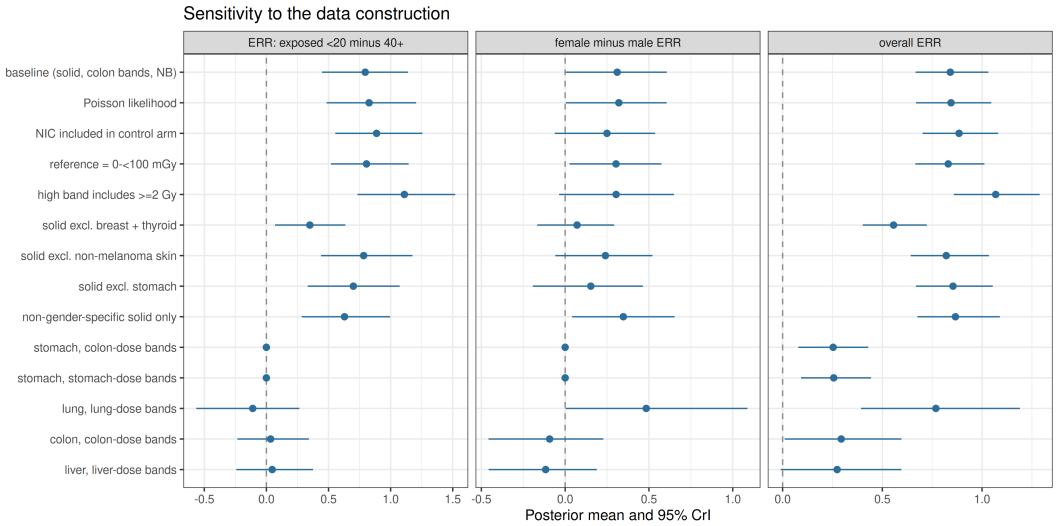


Figure 7: Sensitivity of the featured contrast to the data construction. The average effect and the age-at-exposure gradient are stable across every perturbation except the removal of breast and thyroid, which lowers both.

Not-in-city survivors change nothing. Adding the 1,285 not-in-city cells to the control arm with an indicator in the prognostic forest moves the standardised ERR from 0.495 to 0.501, a change of

1.2% that is well inside the Monte Carlo noise. This reproduces the published conclusion that the dose-response parameter estimates are virtually unchanged when the not-in-city group is excluded, and it answers the design question raised in Section 1.2: the indicator works, and the choice does not matter much either way.

The reference-band and dose-cap definitions are second-order. Widening the reference group to 0–< 100 mGy gives 0.503 against 0.495, and restoring the survivors at or above 2 Gy raises the estimate to 0.588, which is exactly what a dose response predicts, since it raises the exposed arm's mean dose; on a per-Gy scale the two agree. The likelihood does not matter either, with Poisson and negative binomial giving 0.498 and 0.495, consistent with the near-Poisson dispersion posterior of Section 1.5.

Cancer-site composition is the one perturbation that matters, and it settles the sex question. Removing breast and thyroid drops the standardised ERR from 0.495 to 0.405, a fall of 18%, and the female-minus-male ERR difference collapses from 0.311 (0.004, 0.607) to 0.071 (−0.167, 0.292). Removing non-melanoma skin, by contrast, changes almost nothing, giving 0.479 with an age-at-exposure gradient of 0.783 against the baseline 0.795, so non-melanoma skin is not the driver of the aggregate's age-at-exposure shape despite being the site with the steepest age-at-exposure effect in the published heterogeneity tests. The sex effect in the all-solid aggregate is thus carried by the female-only sites, which matches the published finding that the sex difference disappears when the analysis is restricted to non-sex-specific cancers on the EAR scale. The ERR sex difference does not fully disappear in our non-sex-specific fit, where it is 0.347 (0.042, 0.653), which also matches the published analysis, whose ERR sex ratio for non-sex-specific cancers stays at 1.8.

The age-at-exposure gradient survives every data perturbation for all-solid cancer, with a credible interval excluding zero in all nine variants and a magnitude between 0.35 and 1.11. The site-specific results are internally consistent and match the canon where it is sharp. Lung gives a standardised ERR of 0.666 with a credible female-minus-male ERR difference of 0.484 (0.002, 1.088), against published findings of the largest sex effects for lung with female ERRs three to five times the male values. Colon at 0.297 and liver at 0.291 are lower, as expected. For stomach, banding on stomach dose rather than colon dose gives 0.256 against 0.253 with the same complete absence of heterogeneity, so the flat stomach effect surface is not an artefact of dose banding.

1.10 A low-dose city association

The overall 5–< 100 mGy contrast is null, with standardised ERR −0.019 (95% credible interval −0.060, 0.023). Nevertheless, city modifies this contrast, and it is the only dimension that does. The conditional ERR is −0.026 (−0.102, 0.059) in Hiroshima, with a difference from the average of −0.053 (−0.087, −0.022), and 0.209 (0.077, 0.365) in Nagasaki, with a difference of +0.182 (0.078, 0.303); their difference is 0.227 (0.092, 0.378). The Nagasaki effect concentrates in women, with an ERR of 0.249, and in those exposed at ages 10–19 (0.308) and 30–39 (0.203). This city effect is not a consequence of a high-dose city difference, where the two cities do not separate. City is a modifier the canonical model structurally cannot express, since it enters the baseline rate and never the ERR, so this is the clearest instance in the study of a modifier that the established model class cannot represent.

The finding survives a deliberately broad set of checks (Table 12), covering five moderating-forest tree counts, two leaf-concentration multipliers, four propensity specifications, trimming, and the not-in-city group. Across all thirteen specifications the Nagasaki-minus-Hiroshima difference ranges from 0.130 to 0.282 and every 95% credible interval excludes zero, including under the tightest prior the sampler will take. As in the high-dose sweep, the magnitude scales with prior looseness while its existence does not.

Table 12: Robustness of the low-dose (5–< 100 mGy) city difference. Each row is a separate refit. The final column is the posterior Nagasaki-minus-Hiroshima ERR difference with its 95% credible interval; it excludes zero in every specification.

Specification	Hiroshima ERR	Nagasaki ERR	Nagasaki − Hiroshima (95% CrI)
$m_\tau = 10$ (tightest)	−0.029	0.136	0.165 (0.052, 0.269)
$m_\tau = 25$	−0.019	0.184	0.202 (0.072, 0.330)
$m_\tau = 50$ (default)	−0.028	0.199	0.227 (0.092, 0.378)
$m_\tau = 100$	−0.027	0.224	0.251 (0.106, 0.423)
$m_\tau = 200$ (loosest)	−0.017	0.242	0.259 (0.090, 0.449)
$a_0 \times 0.5$	−0.055	0.075	0.130 (0.035, 0.225)
$a_0 \times 2$	−0.012	0.271	0.282 (0.054, 0.545)
Propensity score constant	−0.026	0.191	0.217 (0.077, 0.362)
Propensity score simple logistic	−0.029	0.208	0.237 (excludes 0)
Propensity score rich logistic	−0.025	0.191	0.216 (excludes 0)
Propensity score random forest	−0.034	0.154	0.188 (excludes 0)
Trimmed to $\hat{\pi} \in [0.10, 0.90]$	−0.028	0.199	0.227 (0.092, 0.378)
Not in city added to control arm	−0.023	0.240	0.263 (0.118, 0.437)

An independently fitted adjusted Poisson model with dose-band by city interactions gives the same localisation. The interaction term improves the fit at $p = 0.012$ on a likelihood-ratio test with three degrees of freedom, and the improvement is carried entirely by the low band (Table 13). The finding is therefore not a CountBCF artefact but a feature of the data that a conventional model also sees once it is told to look for it. What CountBCF adds is that it located the modifier without being told.

Table 13: Dose-band by city interaction terms from an adjusted Poisson model. The interaction is confined to the low band. Entries are rate-ratio ratios relative to Hiroshima with 95% confidence intervals.

Interaction term	Rate-ratio ratio (95% CI)	p
5–< 100 mGy × Nagasaki	1.154 (1.054, 1.264)	0.002
100–< 500 mGy × Nagasaki	1.111 (0.976, 1.263)	0.11
500–< 2000 mGy × Nagasaki	1.001 (0.870, 1.151)	0.99

This result requires careful interpretation. The canonical 1958–1998 LSS analysis reported a significant low-dose dose response but did not foreground a city-specific split [Preston et al., 2007]. The later 1958–2009 update found little evidence that city modified the aggregate solid-cancer response [Grant et al., 2017]. These findings make the present low-band association an important question rather than a revision of the established conclusion, and three candidate explanations should be weighed before any reading is adopted.

The first is residual confounding. City is the principal measured imbalance between the reference and low-dose bands. The 5–< 100 mGy band corresponds to roughly 1.8 to 2.5 km from the hypocentre and the reference band to beyond 3 km, and the urban, rural, and socioeconomic composition of those rings differs between the two cities in ways the four available covariates do not capture. The second is dose assignment. The two cities have different shielding histories and different gamma-to-neutron mixtures, so an error in the neutron weighting or in the distal dose imputation would appear as a city difference in risk per weighted Gy, although the neutron fraction

is smallest at these low doses, which argues against that being the whole explanation. The third is multiplicity. This is one positive result among many subgroup examinations, and $p = 0.012$ is not overwhelming. The published comparison of not-in-city baseline rates found a significant Hiroshima effect, at a standardised incidence ratio of 0.93 with $p < 0.001$, but not a Nagasaki one, which hints that city-specific baseline structure in the distal population is real and worth checking further. Reference-group choice and heterogeneity in background incidence are known to affect low-dose and aggregate dose-response inference [Cologne et al., 2019]. The city association is robust to the implemented modelling choices but causally unexplained. It is a target for a dedicated analysis with additional background-rate, geographic, and dosimetric information, not evidence of differential radiosensitivity.

1.11 Propensity score and overlap

The primary propensity score is a logistic regression of band membership on city by sex, natural splines with four degrees of freedom in age at exposure and in attained age, and a city by age-at-exposure interaction. Ground distance is deliberately excluded because it is the assignment mechanism rather than a confounder. Four alternatives were run for comparison: a constant score, a plain additive logistic model, a saturated spline model, and a probability random forest.

Overlap is not strained anywhere. For the featured high-dose contrast the primary score spans 0.117 to 0.752, with no cells below 0.05 or above 0.95, and the same holds for the low and medium contrasts and for every logistic specification. Only the random forest, which overfits with an area under the curve of 0.77 against 0.56 for the logistic models, pushes two cells above 0.95. The score is nearly flat because dose was determined by location and shielding rather than by demography, so the discrimination of the logistic model is barely above chance. Figure 8 displays the distributions.

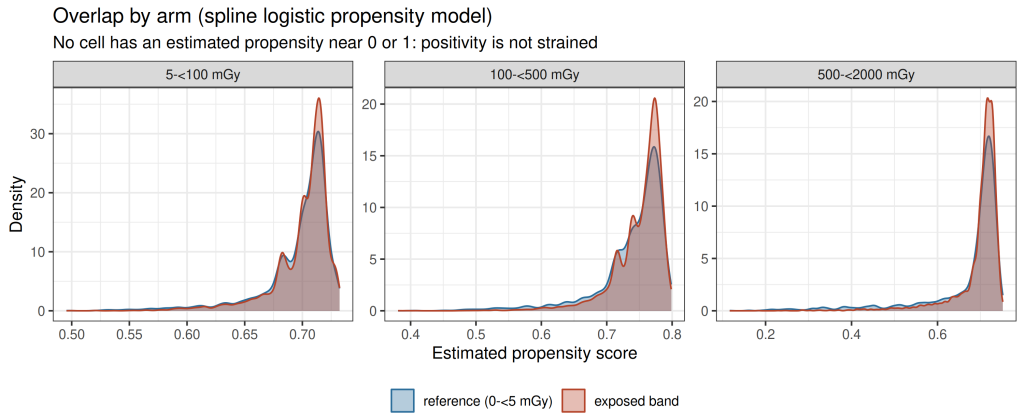


Figure 8: Propensity-score distributions by arm for each dose-band contrast. No cell falls below 0.05 or above 0.95 under the primary specification.

Weighting removes what imbalance is present. Cell-level standardised differences before weighting run from 0.056 to 0.110 for city and reach −0.206 for age at exposure; after inverse-probability weighting with the primary score they are −0.003 to 0.002 for city, −0.002 for age at exposure, −0.003 for attained age, and −0.002 for sex. Trimming is a no-op: because no cell falls outside $[0.10, 0.90]$, trimming at either $[0.05, 0.95]$ or $[0.10, 0.90]$ removes nothing and returns numerically identical results, with all three variants giving 8,249 cells, a standardised ERR of 0.495, and an age-at-exposure ERR difference of 0.795. Capping and overlap weights are therefore unnecessary here.

Table 14: Sensitivity of the featured contrast to the propensity-score specification. Even a constant score reproduces the primary estimate, because the prognostic forest already conditions on city, sex, and both age variables.

Propensity specification	Cells	Standardised ERR	EAR/10^4 PY (95% CrI)	ERR, <20 minus 40+
Constant (no covariates)	8,249	0.497	30.8 (25.1, 36.7)	0.815 (0.473, 1.19)
Simple logistic	8,249	0.500	30.9 (25.1, 36.9)	0.844 (0.494, 1.21)
Spline logistic (primary)	8,249	0.495	30.7 (25.3, 36.5)	0.795 (0.447, 1.14)
Saturated spline logistic	8,249	0.491	30.6 (25.4, 36.2)	0.773 (0.429, 1.13)
Probability random forest	8,249	0.497	30.8 (25.3, 36.4)	0.913 (0.464, 1.39)

The answer does not depend on the score at all (Table 14). Even a constant propensity gives 0.497 against the primary 0.495, because the prognostic forest is already conditioning on city, sex, and both ages, leaving the score little to correct. That is a reassuring finding rather than a disappointing one: the identification here does not hinge on getting the score right.

The featured subgroups are also not in a low-overlap tail. Within the exposed arm of the high-dose contrast, the youngest-at-exposure cells, which carry the large ERR, have the tightest propensity distributions of all, spanning 0.700 to 0.731 for males exposed at ages 0–9 and 0.714 to 0.742 for females, with no cell below 0.05 or above 0.95. The only subgroup with any low-propensity cells is those exposed at age 40 or later, with a minimum of 0.183, and that subgroup sits at the low end of the ERR gradient. Neither the main surface nor the city finding is therefore an artefact of a low-overlap tail.

1.12 Repeated person-years

Each survivor contributes person-years to many cells as they move through attained-age and calendar-period categories, so an independent-cell likelihood should in principle understate the variance. We quantify this with a frequentist cluster bootstrap of the parametric model, resampling the 1,128 survivor cohorts, defined by city, sex, age at exposure, dose category, and distance category, with replacement over 300 replicates (Table 15).

Table 15: Cluster bootstrap of the parametric model over 300 replicates of the 1,128 survivor cohorts. The design effect is the ratio of the bootstrap sampling variance to the model variance (the square of the ratio of the bootstrap standard error to the model standard error).

Dose band	Coefficient	Model SE	Cluster-bootstrap SE	Design effect
Low	−0.0036	0.0205	0.0211	1.06
Medium	0.1112	0.0259	0.0247	0.91
High	0.4261	0.0325	0.0346	1.14

The clustering penalty is at most about 7% on the standard error, and the point estimates do not move. The reason is structural: each cell already aggregates hundreds of survivors, and the exposure is constant within a survivor cohort, so the between-cell correlation induced by repeated person-years is small relative to the Poisson sampling noise in cells that are 74% zeros. This bounds the precision of the independent-cell likelihood without changing any conclusion [Field and Welsh, 2007, Kish, 1995].

We do not report a CountBCF random-intercept fit alongside the bootstrap since its current model version does not handle random-effects. For future developments and generalizations, we

aim to add this feature to the model.

1.13 Low-dose sub-cohorts

Dropkin [2016] re-analysed this same file with generalized additive models restricted to women and to the 0–20, 0–100, and 0–500 mGy sub-cohorts, and reported relative risks at 10 mGy that are ten to forty-five times higher than a linear extrapolation from the full cohort. The same restrictions were run here (Table 16).

Table 16: CountBCF fits on low-dose sub-cohorts. The dose contrast column is the mean weighted-dose separation between the exposed and reference groups.

Sub-cohort	Cells	Cases	Dose contrast (Gy)	Standardised ERR (95% CrI)	EAR/10^4 PY (95% CrI)
5–< 100 mGy, women	4,360	5,474	0.029	0.0006 (−0.061, 0.059)	0.01 (−3.43, 3.13)
5–< 100 mGy, men	4,181	4,535	0.031	−0.047 (−0.110, 0.012)	−3.60 (−8.61, 0.90)
5–< 20 mGy, both sexes	3,839	7,687	0.0096	−0.005 (−0.057, 0.049)	−0.28 (−3.36, 2.90)
5–< 20 mGy, women	1,938	4,239	0.0096	−0.004 (−0.075, 0.065)	−0.25 (−4.09, 3.47)

CountBCF finds no low-dose excess in any of them. This is not a refutation of Dropkin [2016] and should not be presented as one, because the two designs answer different questions. His model fits a smooth dose response in a transformed dose coordinate, the empirical cumulative distribution function of dose, and reads a risk off it at 10 mGy, borrowing strength across the whole low-dose range. Ours contrasts a 5–< 20 mGy band against a 0–< 5 mGy band, a mean dose separation of only 9.6 mGy, with no functional form to borrow from. At that separation the design has essentially no power: the credible interval on the ERR, at roughly ±0.07, corresponds to ±7 per Gy, which cannot discriminate a null from any of the low-dose hypotheses in play. The honest statement is that the categorical-contrast design, adopted for clarity of the causal estimand, is the wrong instrument for the low-dose question, and that a continuous-dose extension of CountBCF is what would be needed to engage with that result directly.

1.14 Excess cases and attributable fractions

Converting the exposed-arm effect back to a case count gives the comparison in Table 17. At medium and high dose the two approaches agree closely, at 372 against 402 excess solid cancers in the 0.5–2 Gy range and 178 against 254 in the 0.1–0.5 Gy range. The gap widens at low dose, where CountBCF returns a null with a slightly negative point estimate while the parametric model assigns 81 excess cases. That is the linear dose-response assumption doing work: the parametric model borrows the slope from the whole 0–2 Gy range and applies it at 30 mGy, whereas the band contrast must find the signal within the band itself. Neither is wrong, but they answer different questions, and the difference illustrates concretely what a nonparametric causal estimator will and will not deliver.

1.15 Caveats and open items

Several limitations qualify the results above and are stated here in full.

The low-dose city difference is statistically robust but causally unexplained. It survives all thirteen specifications and is confirmed by a parametric interaction test, so it is not a modelling artefact, but residual confounding by the urban and rural composition of the distal rings, and multiplicity, both remain live explanations. It should be reported as a robust association and an open question, not as evidence of a city difference in radiosensitivity.

Table 17: Excess cases and attributable fractions from CountBCF, compared with the corresponding published estimates. AF denotes the attributable fraction among the exposed.

Endpoint	Band	Observed cases	CountBCF excess (95% CrI)	CountBCF AF	Published excess	Published AF
All solid	5–< 100 mGy	4,406	−86 (−274, 103)	−2.0%	81	1.8%
	100–< 500 mGy	2,112	178 (68, 286)	8.4%	254	12.0%
	500–< 2000 mGy	1,148	372 (300, 438)	32.4%	402	35.0%
Stomach	5–< 100 mGy	1,230	−58 (−165, 48)	−4.7%	14	1.2%
	100–< 500 mGy	555	0.3 (−57, 60)	0.0%	46	8.2%
	500–< 2000 mGy	269	54 (21, 89)	20.0%	69	25.9%

The high-dose sex effect does not survive the regularisation sweep and should be reported as prior-sensitive. The medium-dose sex effect has not been put through the same sweep and inherits the same doubt until it has been.

The band-contrast design is deliberately blunt at low dose. The 5–< 100 mGy contrast has a mean dose separation of 30 mGy, so nothing about the broader low-dose controversy can be resolved with it.

CountBCF's flat stomach effect surface is a defensible consequence of differential shrinkage on a weak signal, and it is stable across dose banding, but the regularisation sweep was not repeated for stomach. The claim that the data cannot support stomach effect modification therefore rests on the published significance tests plus a single CountBCF fit rather than on a sweep.

Every unit of analysis is a cell rather than a person, so all weighting is by person-years and no individual-level covariate such as smoking, diet, or *Helicobacter pylori* status is available. The stomach results in particular are confounded by unmeasured causes known to matter a great deal for that site in Japan.

The dose is a person-year-weighted cell mean of an adjusted, truncated DS02 organ dose with an assumed 35% dose error already corrected in the RERF release. Dose-measurement error beyond that correction is not modelled.

1.16 Summary

On the full LSS incidence table, CountBCF reproduces the established average solid-cancer dose response and the main age-at-exposure and attained-age gradients without imposing their functional form. Its agreement with the refitted canonical surface is close over most of the observed covariate space, agreeing to four significant figures on the person-year-weighted mean excess rate at high dose, while localized departures at young attained age and short latency identify regions in which parametric extrapolation deserves scrutiny. The age-at-exposure gradient survives all twelve regularisation settings and all nine data perturbations, whereas the high-dose sex effect survives neither and is carried by breast and thyroid. The model appropriately declines to create stomach-cancer heterogeneity unsupported by the site-specific data. Positivity is not strained, the answer does not depend on the propensity specification, and the clustering induced by repeated person-years costs at most about 7% on the standard error. The low-dose city difference is the notable additional empirical finding: it is stable across all thirteen analysis checks and independently visible in a parametric interaction model, yet it remains causally ambiguous. Taken together, these results show that CountBCF can recover established radiation-epidemiology structure, quantify its uncertainty, and expose interpretable departures for further investigation.

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